Questions

Q1.

[*In this question,* ***i*** *and* ***j*** *are horizontal unit vectors.*]

A particle A of mass 0.5 kg is at rest on a smooth horizontal plane.

At time $t = 0$, two forces, $\mathbf{F}_1 = (-3\mathbf{i} + 2\mathbf{j})$ N and $\mathbf{F}_2 = (p\mathbf{i} + q\mathbf{j})$ N, where p and q are constants, are applied to A.

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Given that A moves in the direction of the vector $(\mathbf{i} - 2\mathbf{j})$,

(a) show that $2p + q - 4 = 0$

(4)

Given that $p = 5$

(b) find the speed of A at time $t = 4$ seconds.

(5)

(Total for question = 9 marks)

(**Q06 WME01/01, Oct 2022**)

Q2.

[*In this question,* ***i*** *and* ***j*** *are horizontal unit vectors directed due east and due north respectively and position vectors are given relative to a fixed origin O.*]

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Two ships, A and B, are moving with constant velocities.

The velocity of A is $(3\mathbf{i} + 12\mathbf{j})$ kmh^{-1} and the velocity of B is $(p\mathbf{i} + q\mathbf{j})$ kmh^{-1}

(a) Find the speed of A.

(2)

The ships are modelled as particles.

At 12 noon, A is at the point with position vector $(-9\mathbf{i} + 6\mathbf{j})$ km and B is at the point with position

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vector $(16\mathbf{i} + 6\mathbf{j})$km.

At time t hours after 12 noon,

$$\overrightarrow{AB} = [(25 - 12t)\mathbf{i} - 9t\mathbf{j}]\text{km}$$

(b) Find the value of p and the value of q.

(7)

(c) Find the bearing of A from B when the ships are 15 km apart, giving your answer to the nearest degree.

(7)

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(Total for question = 16 marks)

(**Q08 WME01/01, Oct 2022**)

Q3.

The position vector, $\mathbf{r}$ metres, of a particle P at time t seconds, relative to a fixed origin O, is given by

$$\mathbf{r} = (t - 3)\mathbf{i} + (1 - 2t)\mathbf{j}$$

(a) Find, to the nearest degree, the size of the angle between $\mathbf{r}$ and the vector $\mathbf{j}$, when $t = 2$

(3)

(b) Find the values of t for which the distance of P from O is 2.5 m.

(5)

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(Total for question = 8 marks)

(**Q04 WME01/01, Oct 2021**)

Q4.

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A particle P is moving with constant acceleration $(-4\mathbf{i} + \mathbf{j})\text{ms}^{-2}$

At time $t = 0$, P has velocity $(14\mathbf{i} - 5\mathbf{j})\text{ms}^{-1}$

(a) Find the speed of P at time $t = 2$ seconds.

(3)

(b) Find the size of the angle between the direction of $\mathbf{i}$ and the direction of motion of P at time $t = 2$ seconds.

(3)

At time $t = T$ seconds, P is moving in the direction of vector $(2\mathbf{i} - 3\mathbf{j})$

(c) Find the value of T

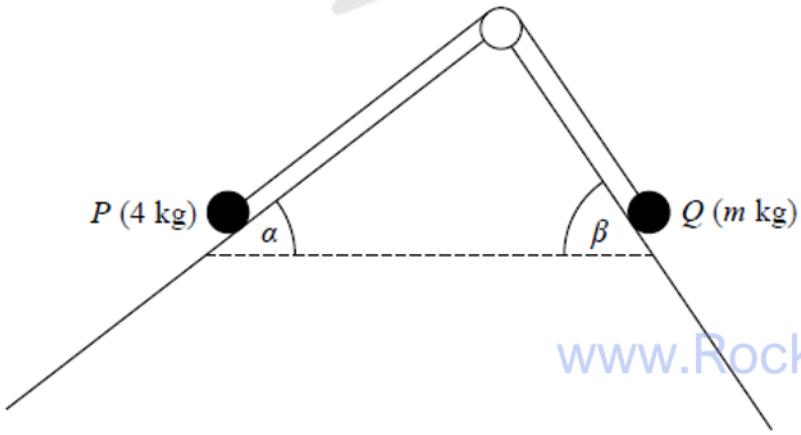
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(4)

(Total for question = 10 marks)

(Q03 WME01/01, Jan 2023)

Q5.



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Figure 3

A particle P of mass 4 kg is attached to one end of a light inextensible string. A particle Q of mass m kg is attached to the other end of the string. The string passes over a small smooth pulley which is fixed at a point on the intersection of two fixed inclined planes. The string lies in a vertical plane that contains a line of greatest slope of each of the two inclined planes. The first plane is inclined to the horizontal at an angle α, where $\tan\alpha = \frac{3}{4}$ and the second plane is inclined to the horizontal at an angle β, where $\tan\beta = \frac{4}{3}$. Particle P is on the first plane and

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particle Q is on the second plane with the string taut, as shown in Figure 3.

The first plane is rough and the coefficient of friction between P and the plane is $\frac{1}{4}$. The second plane is smooth. The system is in limiting equilibrium.

Given that P is on the point of slipping down the first plane,

(a) find the value of m,

(10)

(b) find the magnitude of the force exerted on the pulley by the string,

(4)

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(c) find the direction of the force exerted on the pulley by the string.

(1)

(Total for question = 15 marks)

(**Q07 WME11/01, Specimen papers**)

Q6.

[*In this question* **i** *and* **j** *are horizontal unit vectors directed due east and due north respectively and position vectors are given relative to a fixed origin.*]

At 7 am a ship leaves a port and moves with constant velocity. The position vector of the port is $(-2\mathbf{i} + 9\mathbf{j})$ km.

At 7.36 am the ship is at the point with position vector $(4\mathbf{i} + 6\mathbf{j})$ km.

(a) Show that the velocity of the ship is $(10\mathbf{i} - 5\mathbf{j})$ km h^{-1}

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(2)

(b) Find the position vector of the ship t hours after leaving port.

(2)

At 8.48 am a passenger on the ship notices that a lighthouse is due east of the ship.

At 9 am the same passenger notices that the lighthouse is now north east of the ship.

(c) Find the position vector of the lighthouse.

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(4)

(d) Find the position vector of the ship when it is due south of the lighthouse.

(4)

(Total for question = 12 marks)

(Q08 WME01/01, Oct 2021)

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Q7.

[*In this question* ***i*** *and* ***j*** *are horizontal unit vectors.*]

A particle P of mass 2 kg moves under the action of two forces, $(p\mathbf{i} + q\mathbf{j})$N and $(2q\mathbf{i} + p\mathbf{j})$N, where p and q are constants.

Given that the acceleration of P is $(\mathbf{i} - \mathbf{j})$m s^{-2}

(a) find the value of p and the value of q.

(5)

(b) Find the size of the angle between the direction of the acceleration and the vector $\mathbf{j}$.

(2)

At time $t = 0$, the velocity of P is $(3\mathbf{i} - 4\mathbf{j})$ m s^{-1}

At $t = T$ seconds, P is moving in the direction of the vector $(11\mathbf{i} - 13\mathbf{j})$.

(c) Find the value of T.

(5)

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(Total for question = 12 marks)

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Q8.

[*In this question* ***i*** *and* ***j*** *are horizontal unit vectors directed due east and due north respectively and position vectors are given relative to a fixed origin.*]

A ship A moves with constant velocity $(3\mathbf{i} - 10\mathbf{j})$ km h^{-1}

At time t hours, the position vector of A is $\mathbf{r}$ km.

At time $t = 0$, A is at the point with position vector $(13\mathbf{i} + 5\mathbf{j})$ km.

(a) Find $\mathbf{r}$ in terms of t.

(2)

Another ship B moves with constant velocity $(15\mathbf{i} + 14\mathbf{j})$ km h^{-1}

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At time $t = 0$, B is at the point with position vector $(3\mathbf{i} - 5\mathbf{j})$ km.

(b) Show that, at time t hours,

$$\overrightarrow{AB} = [(12t - 10)\mathbf{i} + (24t - 10)\mathbf{j}]\text{km}$$

(4)

Given that the two ships do not change course,

(c) find the shortest distance between the two ships,

(6)

(d) find the bearing of ship B from ship A when the ships are closest.

(2)

(Total for question = 14 marks)

(Q08 WME01/01, Jan 2022)

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Q9.

A particle P is moving with constant acceleration.

At time $t = 1$ second, P has velocity $(-\mathbf{i} + 4\mathbf{j})$ m s^{-1}

At time $t = 4$ seconds, P has velocity $(5\mathbf{i} - 8\mathbf{j})$ m s^{-1}

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Find the speed of P at time $t = 3.5$ seconds.

(Total for question = 6 marks)

(Q06 WME01/01, June 2022)

Q10.

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[*In this question,* **i** and **j** are horizontal unit vectors directed due east and due north respectively and position vectors are given relative to a fixed origin O.]

Two boats, P and Q, are moving with constant velocities.

The velocity of P is $15\mathbf{i}$ m s^{-1} and the velocity of Q is $(20\mathbf{i} - 20\mathbf{j})$ m s^{-1}

(a) Find the direction in which Q is travelling, giving your answer as a bearing.

(2)

The boats are modelled as particles.

At time $t = 0$, P is at the origin O and Q is at the point with position vector $200\mathbf{j}$ m.

At time t seconds, the position vector of P is $\mathbf{p}$ m and the position vector of Q is $\mathbf{q}$ m.

(b) Show that

$$\overrightarrow{PQ} = [5t\mathbf{i} + (200 - 20t)\mathbf{j}] \text{ m}$$

(5)

(c) Find the bearing of P from Q when $t = 10$

(2)

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(d) Find the distance between P and Q when Q is north east of P

(5)

(e) Find the times when P and Q are 200 m apart.

(3)

(Total for question = 17 marks)

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(**Q08 WME01/01, June 2022**)

Q11.

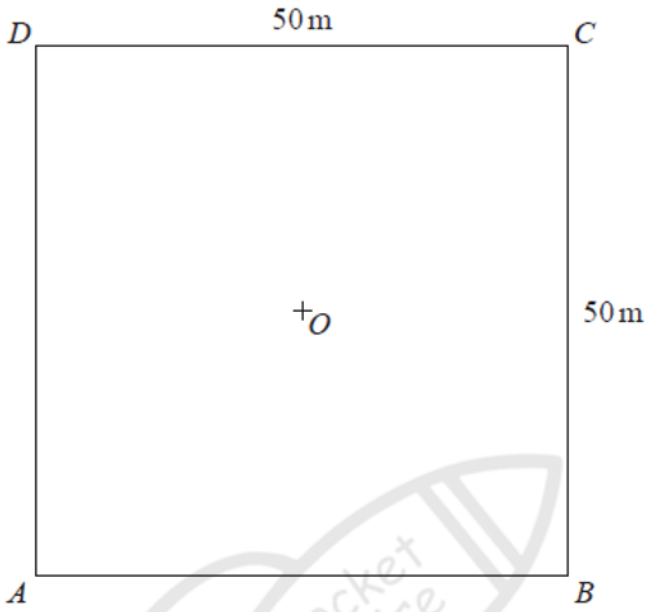


Figure 5

A square floor space *ABCD*, with centre *O*, is modelled as a flat horizontal surface measuring 50 m by 50 m, as shown in Figure 5.

The horizontal unit vectors **i** and **j** are in the direction of $\overrightarrow{AB}$ and $\overrightarrow{AD}$ respectively.

All position vectors are given relative to *O*.

A small robot *R* is programmed to travel across the floor at a constant velocity.

- At time $t = 0$, *R* is at the point with position vector $(-2\mathbf{i} + \mathbf{j})$ m
- At time $t = 11$ s , *R* is at the point with position vector $(9\mathbf{i} + 23\mathbf{j})$ m
- At time *t* seconds, the position vector of *R* is **r** metres

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(a) Find, in terms of *t*, **i** and **j**, an expression for **r**

(3)

A second robot *S* is at the point *C*.

- At time $t = 0$, *S* leaves *C* and moves with constant velocity $(-\mathbf{i} - \mathbf{j})$ m s^{-1}
- At time *t* seconds, the position vector of *S* is s metres

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(b) Write down, in terms of t, **i** and **j**, an expression for s

(1)

(c) Show that

$$\overrightarrow{SR} = [(2t - 27)\mathbf{i} + (3t - 24)\mathbf{j}] \text{ m}$$

(2)

(d) Find the time when the distance between R and S is a minimum.

(3)

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(Total for question = 9 marks)

(Q08 WME01/01, June 2023)

Q12.

Two forces, $\mathbf{F}_1$ and $\mathbf{F}_2$, act on a particle A.

$\mathbf{F}_1 = (2\mathbf{i} - 3\mathbf{j})$ N and $\mathbf{F}_2 = (p\mathbf{i} + q\mathbf{j})$ N, where p and q are constants.

Given that the resultant of $\mathbf{F}_1$ and $\mathbf{F}_2$ is parallel to $(\mathbf{i} + 2\mathbf{j})$,

(a) show that $2p - q + 7 = 0$

(5)

Given that $q = 11$ and that the mass of A is 2 kg, and that $\mathbf{F}_1$ and $\mathbf{F}_2$ are the only forces acting on A,

(b) find the magnitude of the acceleration of A.

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(5)

(Total for question = 10 marks)

(Q05 WME11/01, Specimen papers)

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